

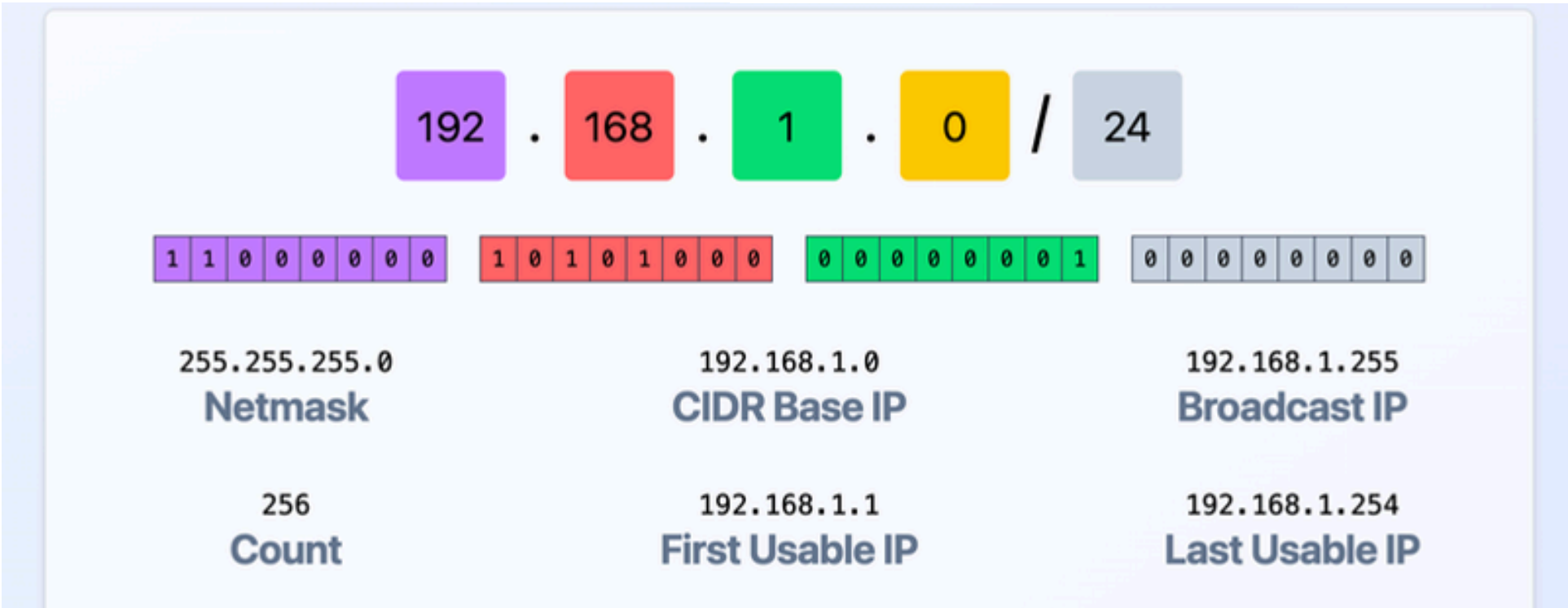
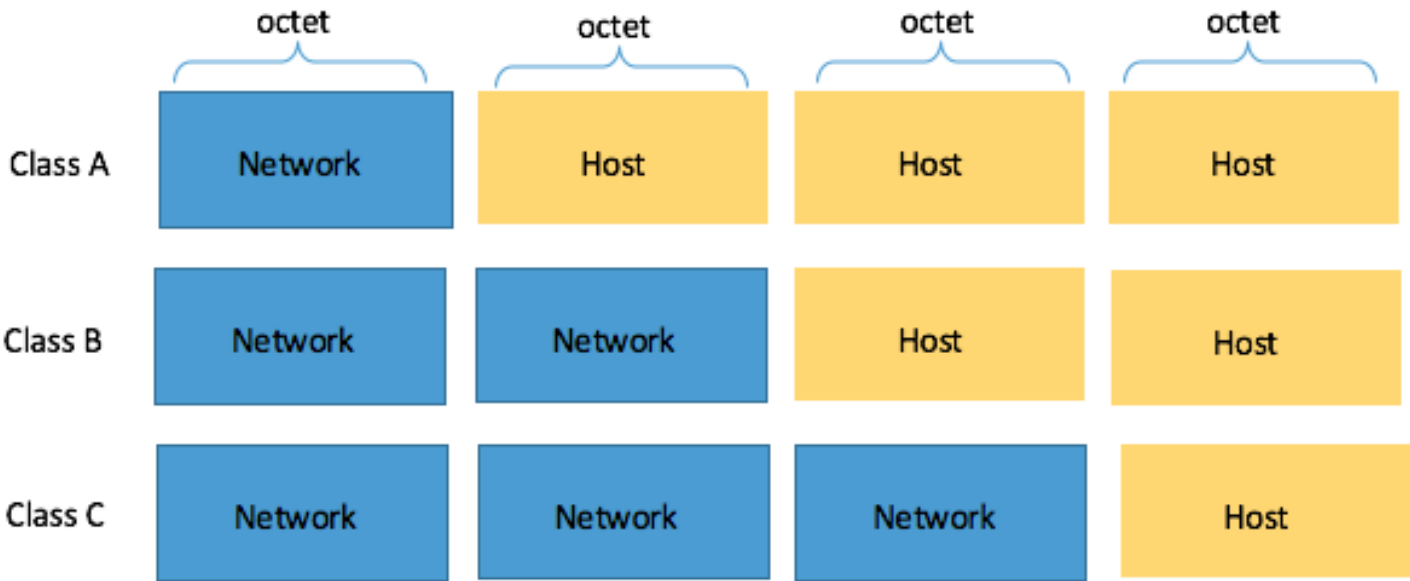
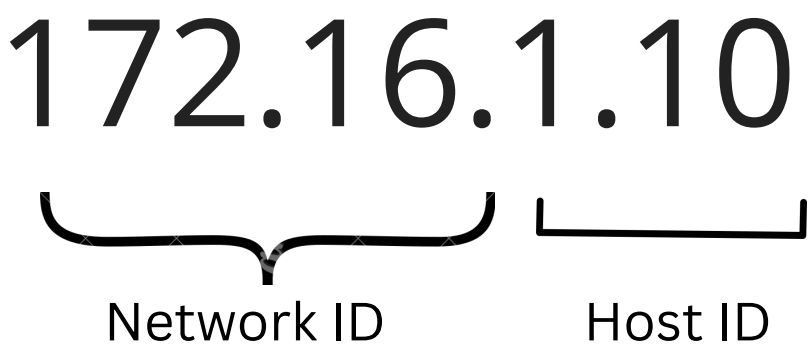
Subnetting in Practice – and Designing an AWS VPC

Computer Networks

Join the lecture online on your dashboard

Recap

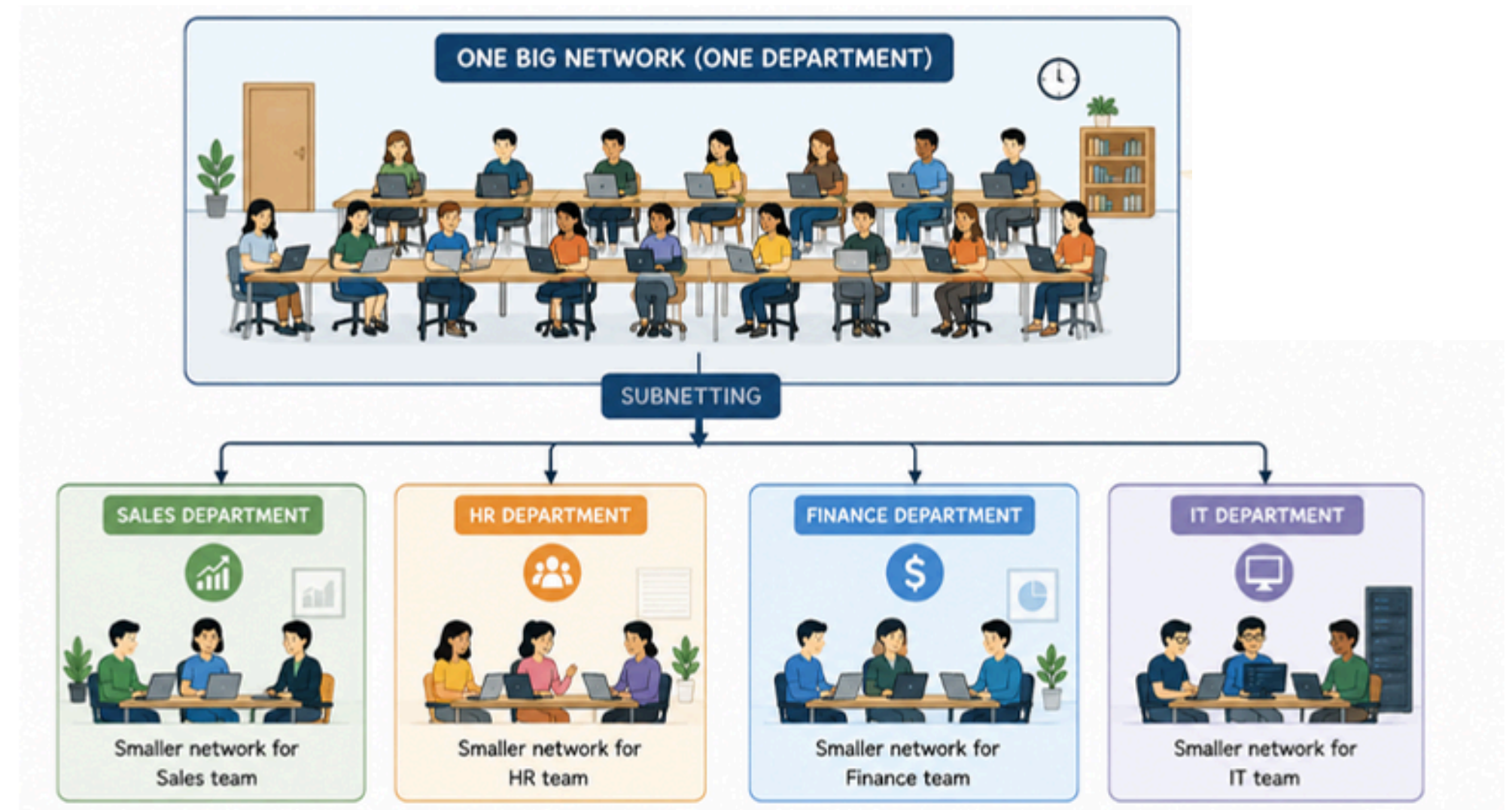
IP Address



**One large network becomes difficult to
manage**

Subnetting = Dividing One Network

Subnetting means splitting one big network into smaller networks.



Borrowing Host Bits

Now borrow k host bits for subnetting.

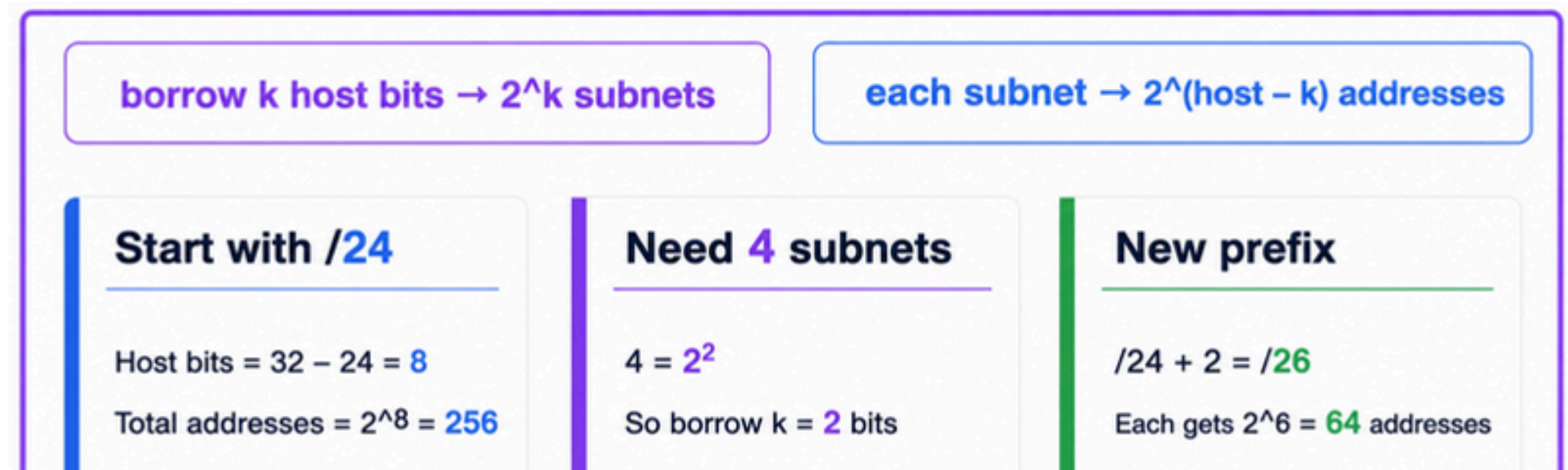
We do subnetting by borrowing bits from the host part.

CIDR Block has two parts:

Network Bits + Host Bits

Example: 10.0.0.0/24

- /24 means 24 network bits
- $32 - 24 = 8$ host bits



Magic Number Shortcut

Worked example: 10.0.0.0/24 → four /26s



Block size = 64

Network addresses jump by 64: .0, .64, .128, .192

For /26, subnet mask is: 255.255.255.192

Magic number: $256 - 192 = 64$

So subnet blocks jump by 64: .0, .64, .128, .192

Network, Usable Range, Broadcast

- Four /26 Subnets from 10.0.0.0/24

Subnet	Network	Usable Range	Broadcast
1	10.0.0.0/26	10.0.0.1 – 10.0.0.62	10.0.0.63
2	10.0.0.64/26	10.0.0.65 – 10.0.0.126	10.0.0.127
3	10.0.0.128/26	10.0.0.129 – 10.0.0.190	10.0.0.191
4	10.0.0.192/26	10.0.0.193 – 10.0.0.254	10.0.0.255

Not Every Subnet Should Face the Internet

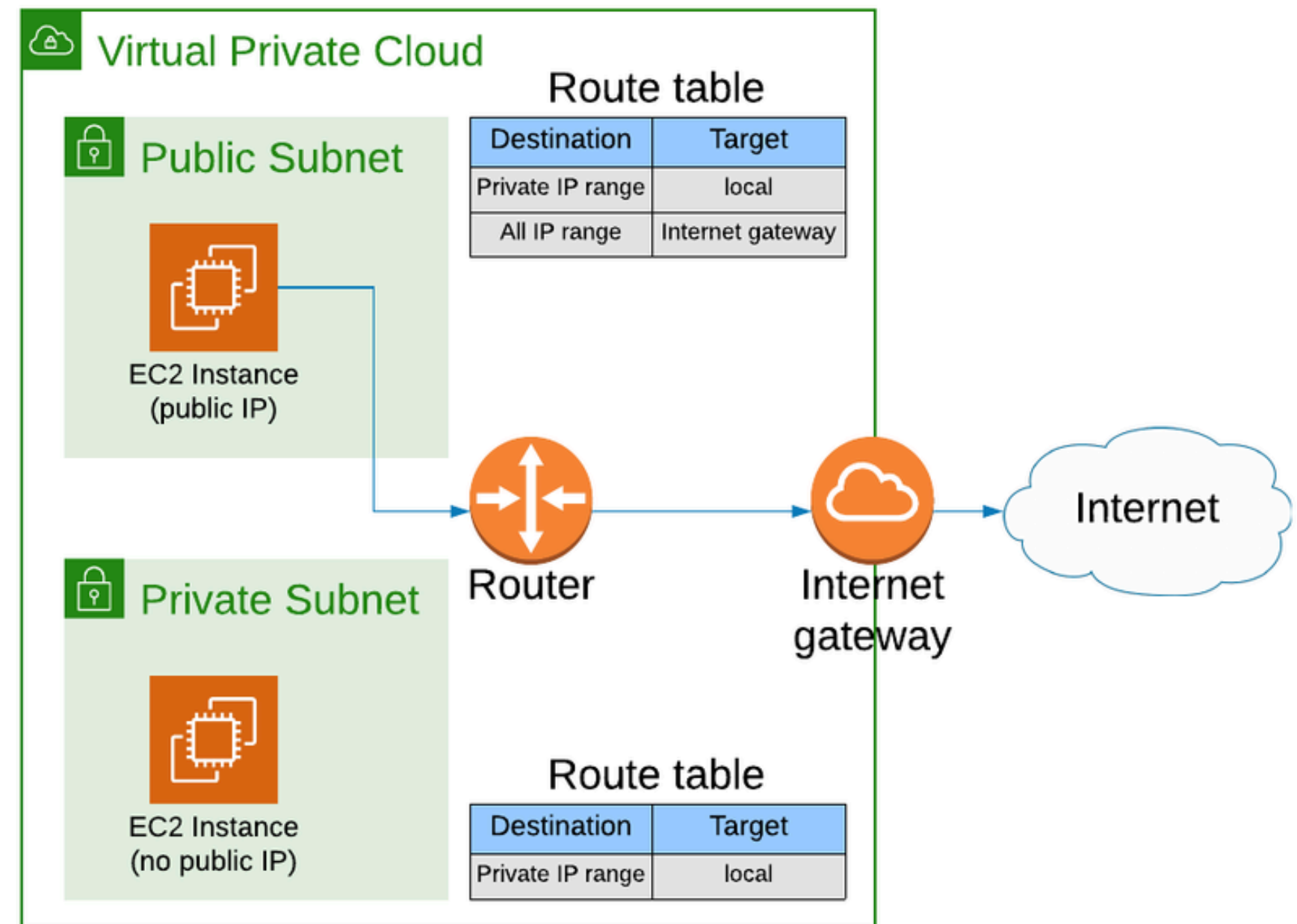
Public vs Private Subnets

In a VPC, subnets can be public or private.

A subnet become public because its route table has a path to the Internet Gateway.

A subnet is private if its route table does not send:

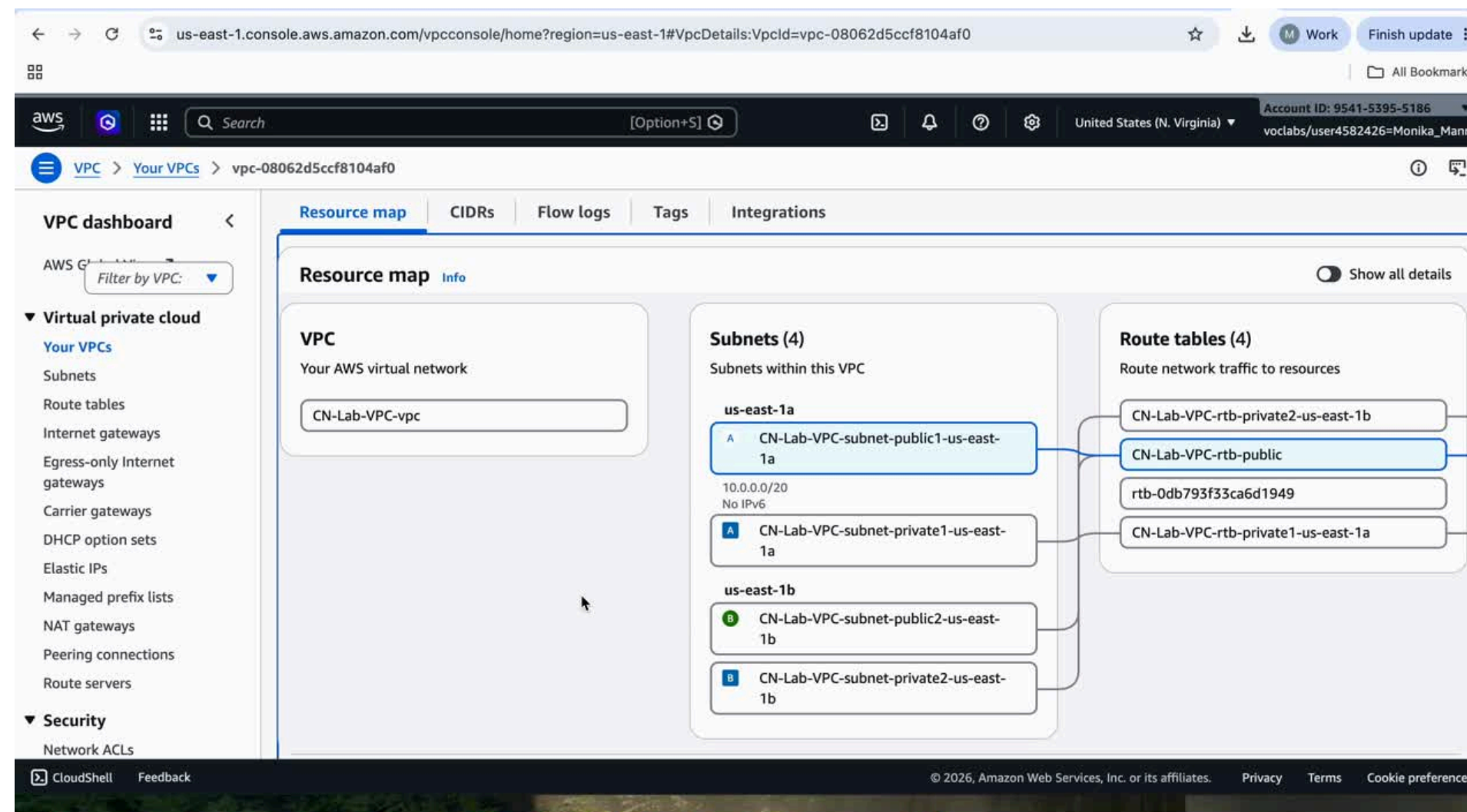
0.0.0.0/0 → Internet Gateway



What Makes a Subnet Public or Private

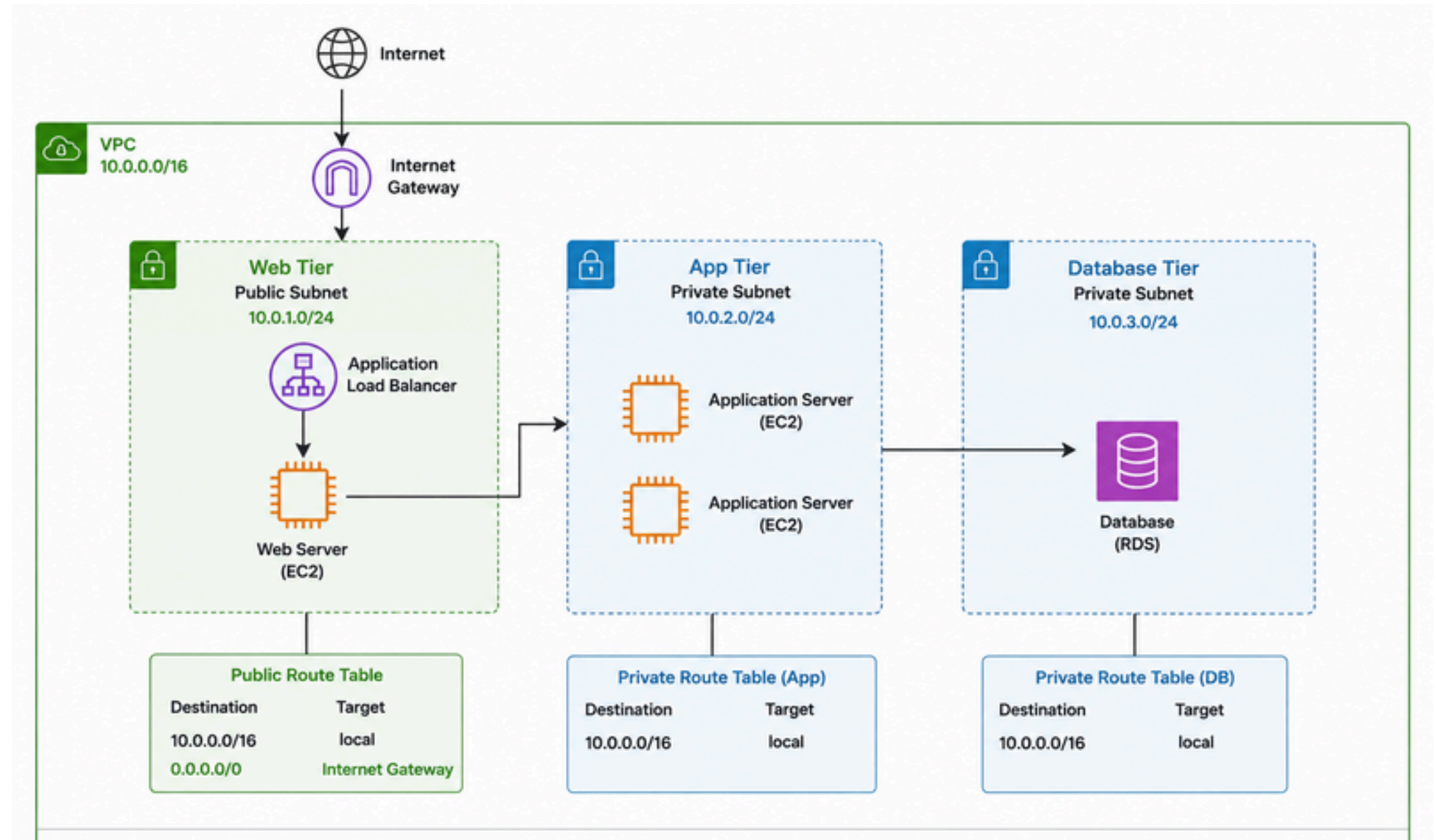
Public Subnet → Route Table → IGW → Internet

Private Subnet → Route Table → NAT → Internet



3-Tier Architecture

- **Web tier:** Public subnet
- **App tier:** Private subnet
- **Database tier:** Private subnet



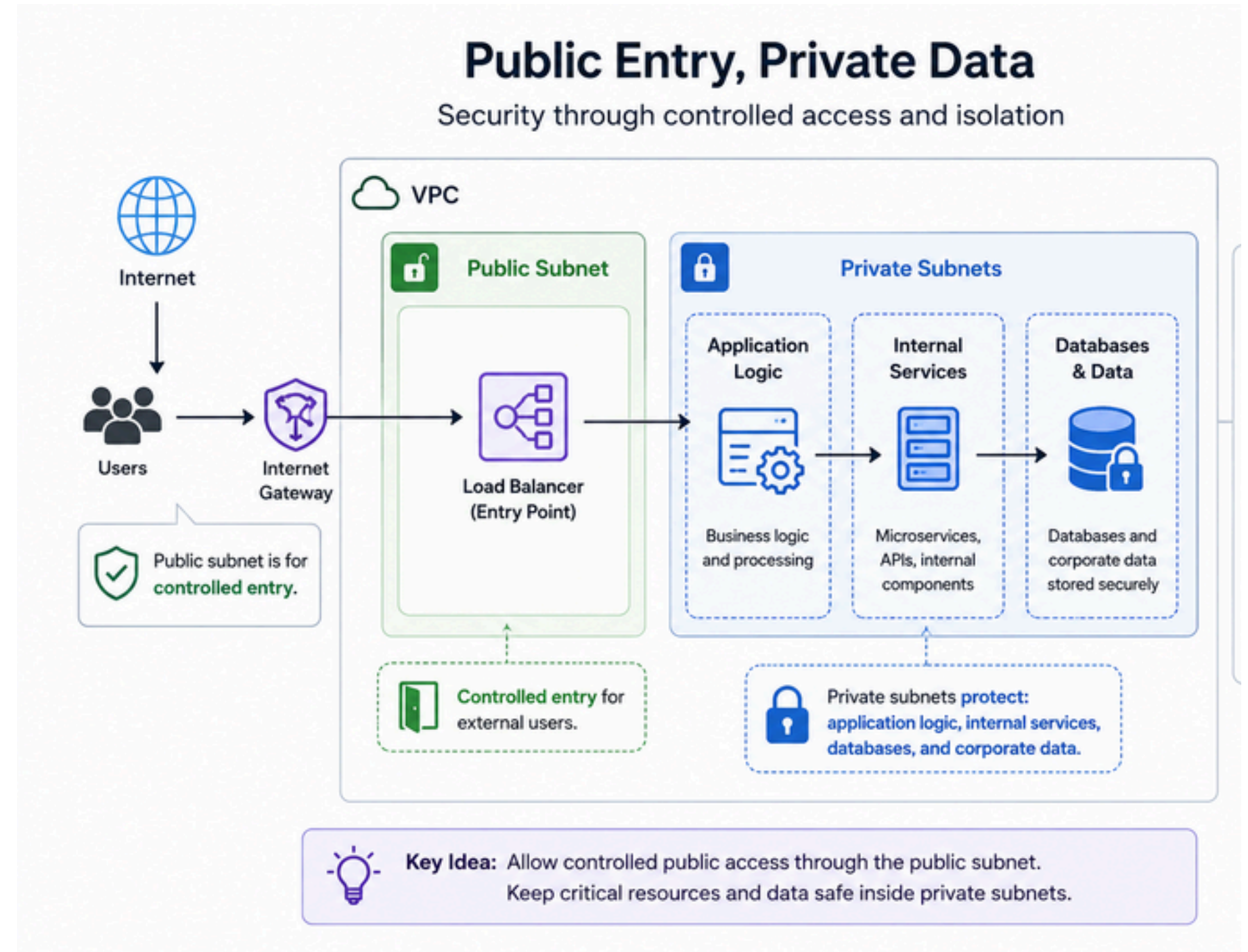
Why This Keeps Corporate Data Safe

Public Entry, Private Data

Public subnet is for controlled entry.

Private subnets protect:

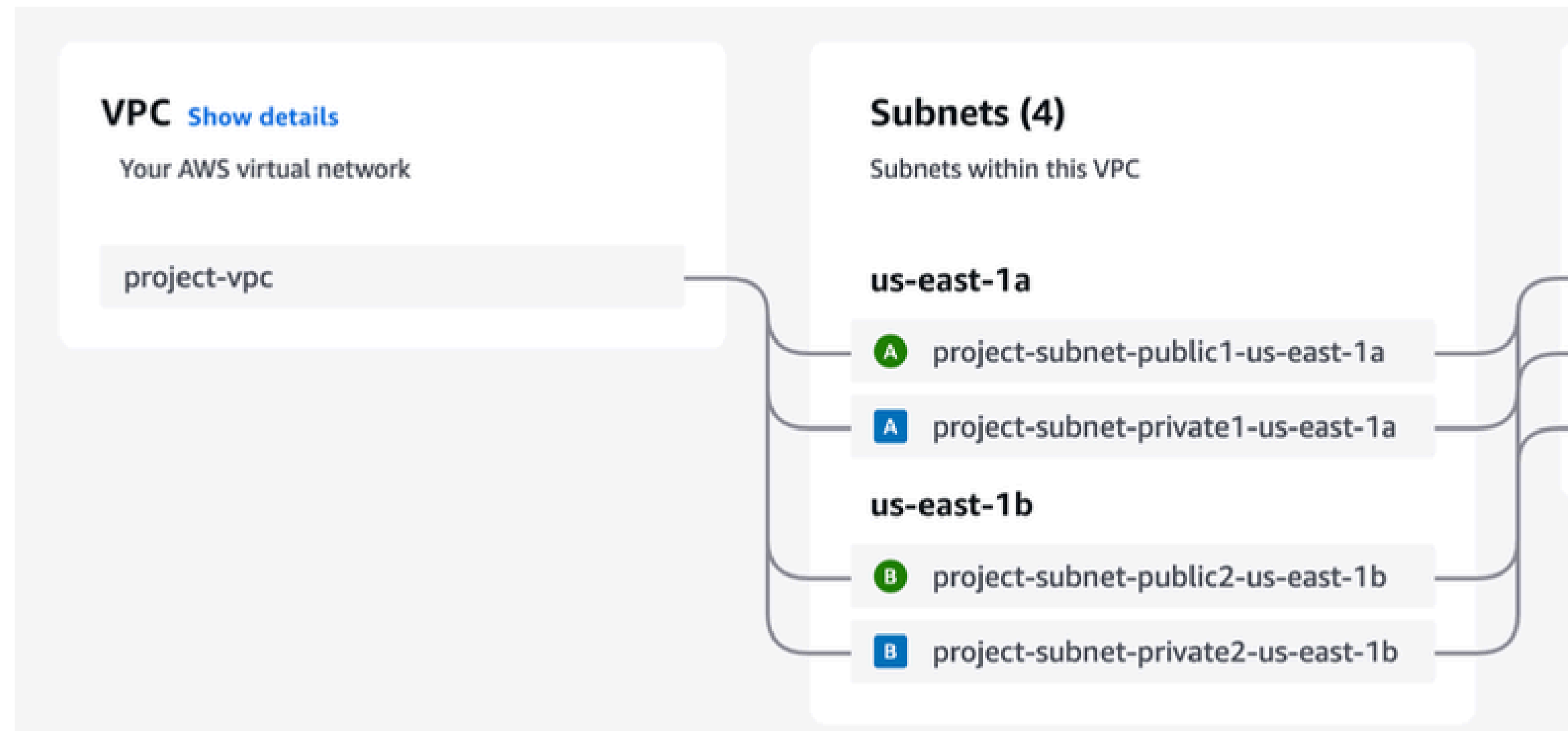
- application logic
- internal services
- databases
- corporate data



Duplicate Subnets Across Multiple AZs

- For availability, repeat the same tiers across at least 2 Availability Zones.

Example:



The AWS Reality — 5 Reserved IPs & Constraints

The AWS Reality: Not All IPs Are Usable

- In normal subnetting, we calculate total addresses.
- In AWS, every subnet loses 5 reserved IP addresses. So usable IP count is always:
Total IPs - 5

IP Address	Reserved For
10.0.0.0	Network address
10.0.0.1	VPC router
10.0.0.2	DNS
10.0.0.3	Future use
10.0.0.255	Broadcast

Traditional thinking:
/24 → 254 usable hosts
AWS reality:
/24 → 251 usable IPs

Demo: AWS Shows Available IPs

Subnets (1/10) [Info](#)

Last updated 20 minutes ago

Actions

Create subnet

< 1 >

	Name	Subnet ID	State	VPC	Block Public...	
☒	CN-Lab-VPC-subnet-private2-us-east-1b	subnet-0751527e050159830	Available	vpc-08062d5ccf8104af0 CN-L...	Off	1
☐	CN-Lab-VPC-subnet-public1-us-east-1a	subnet-05d488676845659e3	Available	vpc-08062d5ccf8104af0 CN-L...	Off	1
☐	-	subnet-0db376f5974b502b7	Available	vpc-0f34a71fce73316ff	Off	1
☐	-	subnet-0921c3e5132120302	Available	vpc-0f34a71fce73316ff	Off	1
☐	-	subnet-0a178c5d3700e46d8	Available	vpc-0f34a71fce73316ff	Off	1

subnet-0751527e050159830 / CN-Lab-VPC-subnet-private2-us-east-1b

Subnet ID
 subnet-0751527e050159830

IPv4 CIDR
 10.0.144.0/20

Availability Zone
 use1-az6 (us-east-1b)

Network ACL
[acl-0dac011c03a5bfa8d](#)

Subnet ARN
 arn:aws:ec2:us-east-1:954153955186:subnet/subnet-0751527e050159830

Available IPv4 addresses
 4091

Network border group
 us-east-1

State
 Available

IPv6 CIDR
-

VPC
[vpc-08062d5ccf8104af0](#) | CN-Lab-VPC-vpc

Auto-assign public IPv4 address
...

Block Public Access
 Off

IPv6 CIDR association ID
-

Route table
[rtb-05688e9489fe1712a](#) | CN-Lab-VPC-rtb-private2-us-east-1b

Auto-assign IPv6 address
...

AWS CIDR Size Limits

AWS allows IPv4 CIDR block sizes from: /16 to /28

- Largest block: /16 and Smallest block: /28
- You cannot create very tiny subnets like /29 or /30

☰ [VPC](#) > [Subnets](#) > Create subnet

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

my-subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1a) ▼

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16 ▼

IPv4 subnet CIDR block

10.0.32.0/29 8 IPs

< > ^ ▼

✖ IPv4 block sizes must be between a /16 netmask and /28 netmask.

AWS CIDR Size Limits

VPC > Subnets > Create subnet

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10.0.0.0/16

IPv4 subnet CIDR block

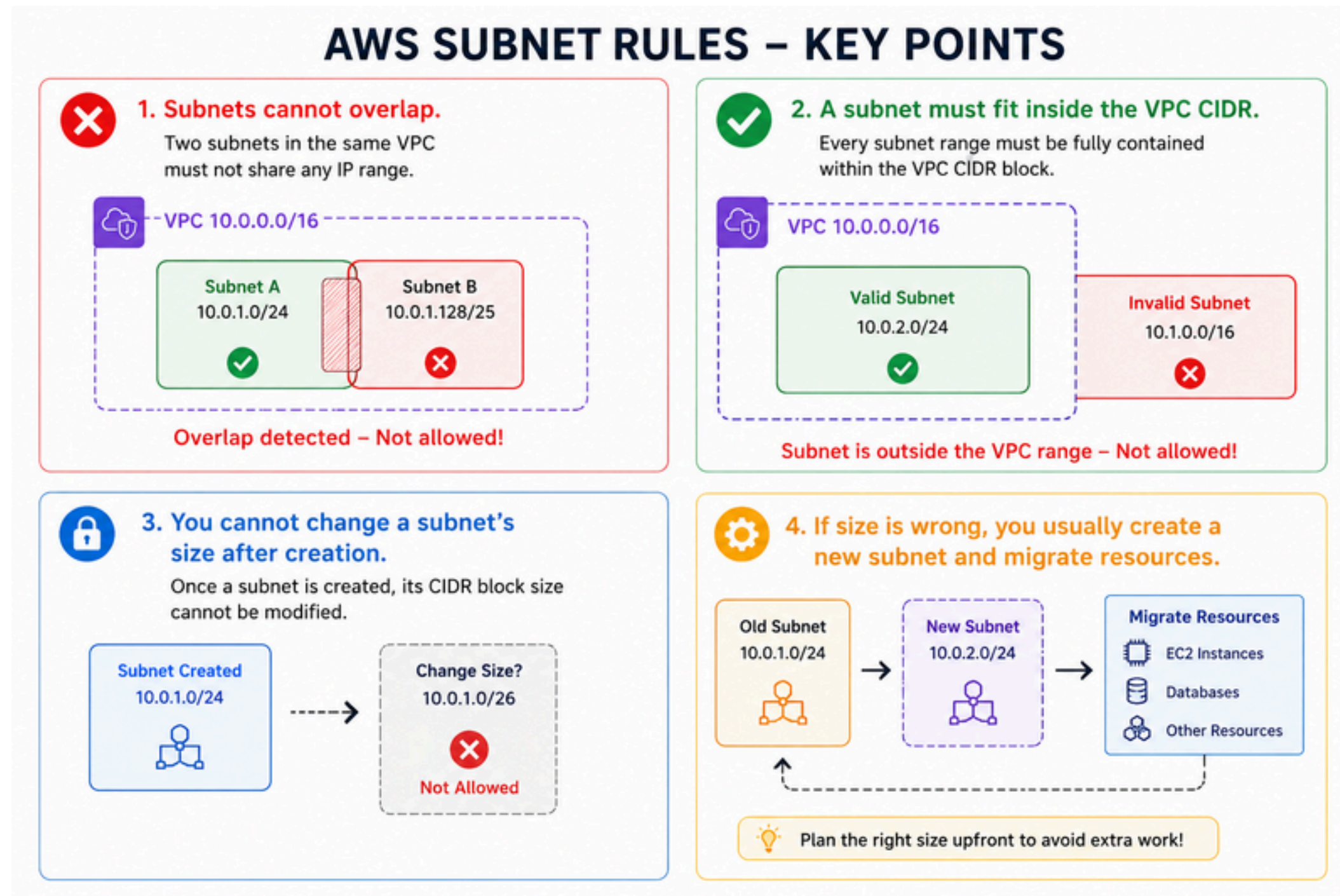
10.0.32.0/26

64 IPs

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Subnet Design Constraints



Please fill the feedback form.

Thank You

